GEK-83784

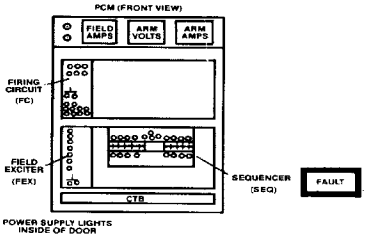
DS3827 SILTRON

Power Conversion Module

Troubleshooting Charts

INSTRUCTIONS

HEAVY LINED BOXES ARE "MILESTONE" CHECKS
EX: WHEN DRIVE WILL NOT RUN AND "M" PICKS UP
PROBLEM IS BELOW THAT MILESTONE BOX
HEAVY LINES MEANS COMMON FLOW.USE TO ELIMINATE POSSIBLE PROBLEM AREAS.
EX: WHEN DRIVE WILL NOT RUN, PUSH JOG FIRST:
IF THIS WORKS ALL COMMON ELEMENTS ARE OK



FAULT

DRIVE WILL NOT MOVE

PS CR19 — IS YELLOW IMOK LIGHT ON, POWER SUPPLY ON — NO → 5V POWER SUPPLY PROBLEM SEE SUBROUT 1 — YES

SEQ IMOK — IS YELLOW IMOK LIGHT ON SEQ. BD ON — NO → REPLACE SEQUENCER BOARD — YES

FC IMOK — IS YELLOW IMOK LIGHT ON FIRING CKT. BD ON — NO → SEQ ORST — IS ORST LIGHT OFF — NO / YES → REPLACE SEQ BOARD OR FIR CKT. BD. — YES

SEQ F — IS FAULT LIGHT OFF — NO → IS FAULT LIGHT FLASHING — NO → THIS FAULT STOP WAS NOT CAUSED BY THIS DRIVE. PUSH OP. ST. RESET OR NORMAL STOP — NO → PULL JC CABLE ON SEQ. DRIVE IF STILL FAULTED PROCEED TO FAULT PROBLEM SH 2

IS FAULT LIGHT FLASHING — YES → SEE SH. 2

IS FAULT LIGHT OFF — YES → YES (RUN OR THREAD/SLOW) / YES (JOG REVERSE) / YES (JOG FORWARD)

SEQ RR — IS "READY TO RUN" LIGHT ON — NO → IS THE CASCADE SWITCH ON THE REG. BD IN CASC POSITION — NO → ARE NONE OF THE INPUT LIGHTS ON — NO → SEQ QSTP — IS EXTERNAL QUICK STOP LIGHT ON — NO → INPUT PROBL SEE SUB. ROUTINE 2 (SHEET 3)

IS THE CASCADE SWITCH ON THE REG. BD IN CASC POSITION — YES → MOVE TO 'OFF' POSITION AND TRY AGAIN

ARE NONE OF THE INPUT LIGHTS ON — YES → POSSIBLY 105V POW SUP FAILURE SEE SUBROUT 1

IS EXTERNAL QUICK STOP LIGHT ON — YES → SEQ RPER — IS RUN PERMISSIVE LIGHT ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

IS RUN PERMISSIVE LIGHT ON — YES → SEQ SPER — IS START PERMISSIVE LIGHT ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

IS START PERMISSIVE LIGHT ON — YES → IF YOU UNPLUG JC CABLE DOES READY TO RUN LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

IF YOU UNPLUG JC CABLE DOES READY TO RUN LIGHT COME ON — YES → DRIVE IS STOPPED BY ANOTHER DRIVE

SEQ NSTP — IS "NORMAL STOP" LIGHT ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 SHEET 3

IS "NORMAL STOP" LIGHT ON — YES → YES (THREAD)

RUN | **THREAD** | **JOG FWD** | **JOG REV**

SEQ STRT — WHEN PUSHING START/RUN DOES START LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

SEQ THD — WHEN PUSHING THREAD/SLOW DOES THREAD LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

SEQ JF — WHEN PUSHING JOG FWD DOES JOG FWD LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 (SHEET 3)

SEQ JR — WHEN PUSHING JOG REV DOES JOG REV LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 SHEET 3

SEQ RUN — DOES "RUNNING" LIGHT STAY ON — NO

SEQ ME — DOES ME LIGHT COME ON — NO → IS THE CASCADE SWITCH ON THE REG. BD IN CASC POSITION — NO → SEE ROUTINE 1 FOR 28V PS FAILURE — NO → REPLACE SEQUENCER AND/OR I/O BD.

IS THE CASCADE SWITCH ON THE REG. BD IN CASC POSITION — YES → TRY AGAIN

DOES M-CONTACTOR PICK UP — NO → CAN YOU MEASURE VOLT AT COIL — NO → CAN YOU MEASURE VOLT AT CTB 51 — NO → CHECK EXTERNAL WIRING INTO CTB 51

CAN YOU MEASURE VOLT AT COIL — YES → REPLACE CONTACTOR

CAN YOU MEASURE VOLT AT CTB 51 — YES → REPLACE SEQUENCER BOARD

DOES M-CONTACTOR PICK UP — YES

SEQ M — DOES M LIGHT COME ON — NO → INPUT PROBL SEE SUB-ROUTINE 2 SHEET 3

DOES M LIGHT COME ON — YES → YES (RUN)

M-CONTACTOR PICKS UP BUT DRIVE WILL NOT RUN

SEQ RREF — DOES "RUN REFERENCE" LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

SEQ TREF — DOES "THREAD/SLOW REF" LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

SEQ JFR — DOES JOG FWD REF. LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

SEQ JRR — DOES JOG REV. REF. LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

YES (THREAD) / YES (JOG FWD.) / YES (JOG REV.)

J41R IN — YES

J40R IN

SEQ RAMP REL — DOES "RAMP RELEASE" LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

YES (RUN OR THREAD)

SEQ CRA REL — DOES CURR REGULATOR SUICIDE LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

SEQ VRA REL — DOES VOLT REGULATOR SUICIDE LIGHT COME ON — NO → REPLACE SEQUENCER BOARD

SEQ SRA REL — DOES SPEED REGULATOR SUICIDE LIGHT COME ON — NO → SEQ FTFB — IS FAULT MASK LIGHT ON — NO → REPLACE SEQUENCER BD

IS FAULT MASK LIGHT ON — YES → SEQ FLT MASK — IS EXTERNAL FAULT MASK LIGHT ON — NO → REPLACE SEQ. BD.

IS EXTERNAL FAULT MASK LIGHT ON — YES → INPUT PROBL SEE SUB-ROUTINE 2

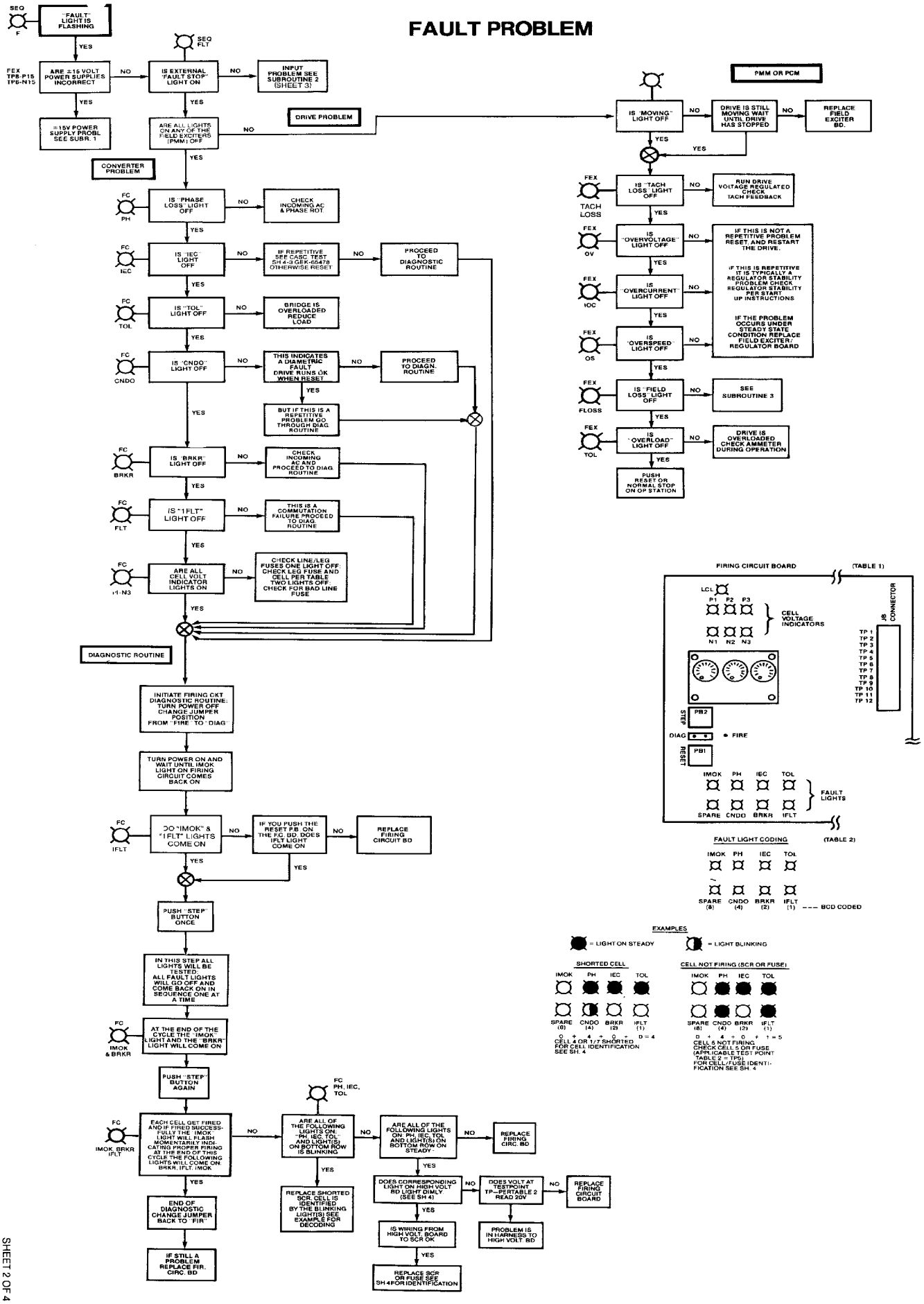
DOES SPEED REGULATOR SUICIDE LIGHT COME ON — YES → FEX TP6-P15 TP6-N15 — ARE ±15 VOLT POWER SUPPLIES OK CHECK TESTP — NO → POWER SUPPLY PROBL SEE SUBR. 1

ARE ±15 VOLT POWER SUPPLIES OK CHECK TESTP — YES → SEE CASCADE TESTING SH. 4-3 GEK-65479

IS REVERSING LIGHT ON — NO → REPLACE SEQ. BD.

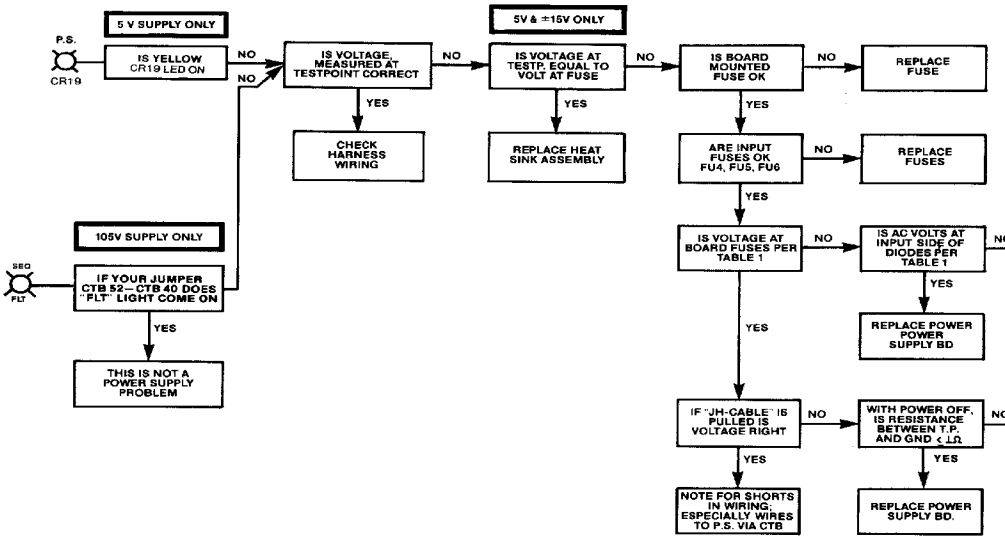
IS REVERSING LIGHT ON — YES

FAULT PROBLEM



SUBROUTINE 1

POWER SUPPLY PROBLEM: 5VOLT +15VOLT −15VOLT 28VOLT 105VOLT 115VAC

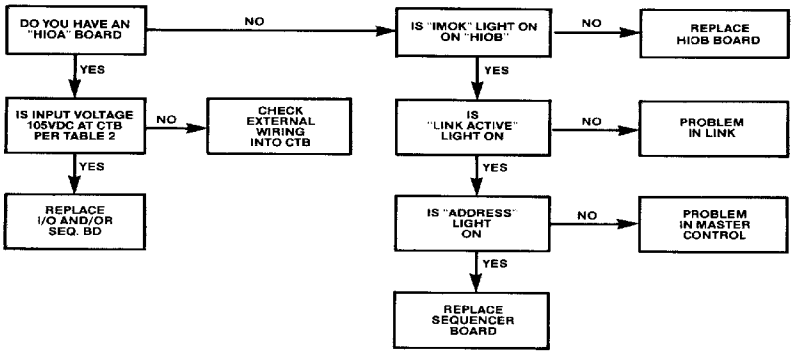


POWER SUPPLY VOLT	VOLT AT FUSE	TEST POINT	VOLTS AT DIODE
5V	FU2	5V ON P.S.	8V AT CR11-CR16
+15V	FU3	+15V ON P.S.	17.5V AT CR 21-26
−15V	FU4	−15V ON P.S.	17.5V AT CR21-26
28V	FU1	28V ON P.S.	20V AT CR1-6
105V	FU5	CTB 52 105V	NA
115VAC	FU6 NIA	NA	NA

TABLE 1
ALL VOLTAGES REFERENCED TO ACOM

SUBROUTINE 2

INPUT PROBLEM



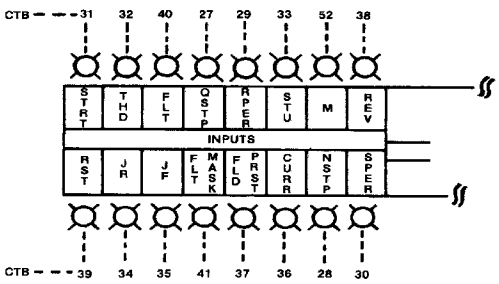
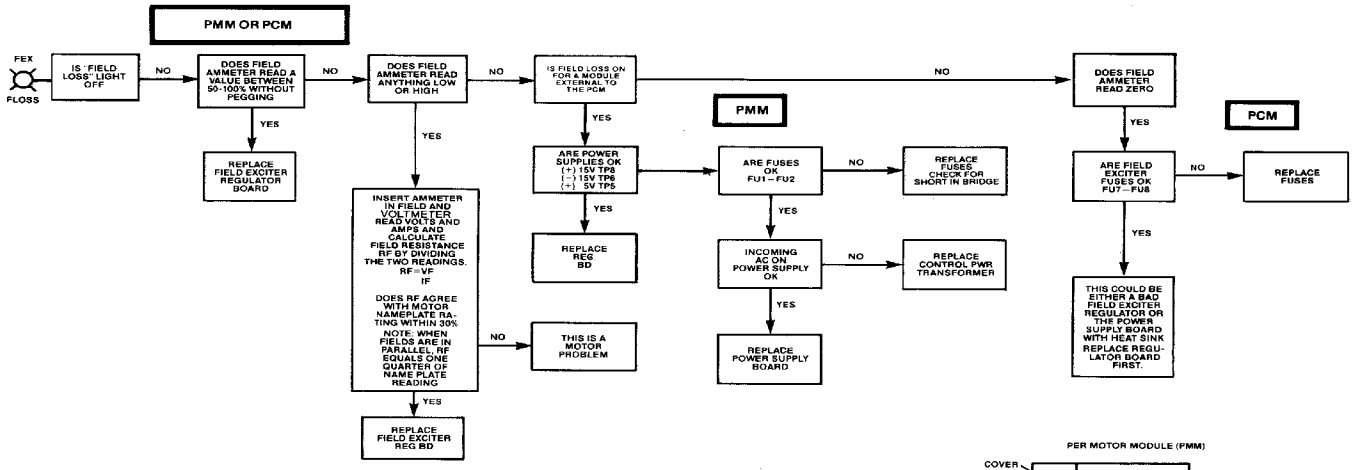
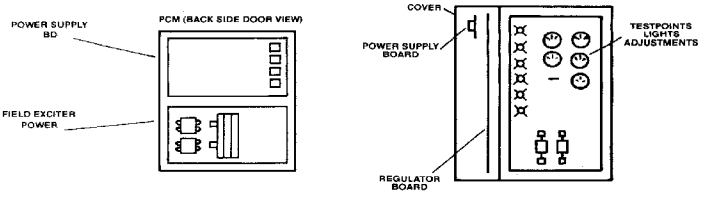


TABLE 2

SUBROUTINE 3

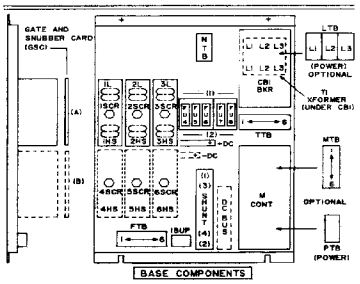
FIELD LOSS PROBLEM

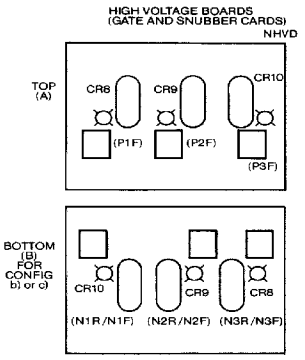




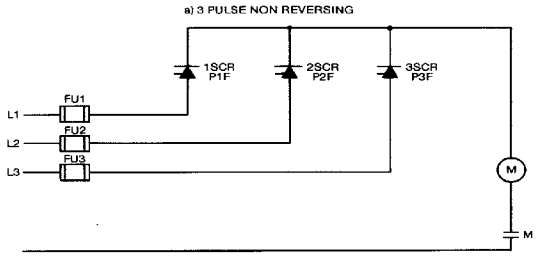
CELL/FUSE IDENTIFICATION

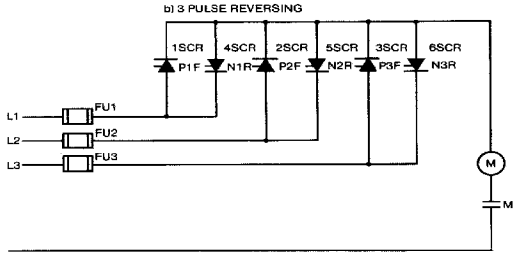
LOW HORSEPOWER

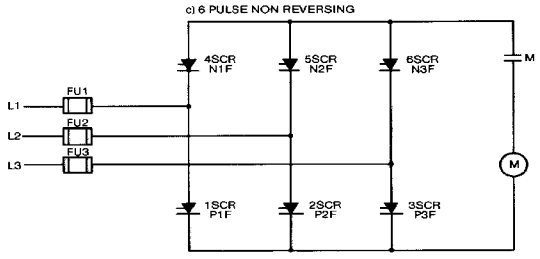




CONFIGURATIONS:







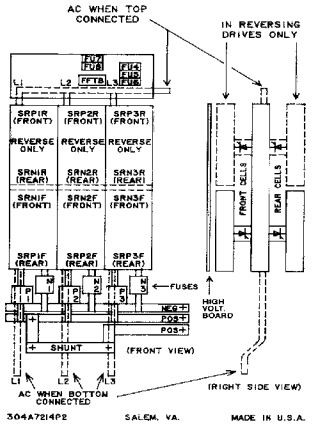
CELL	NOMEN CLAT.	CONFIG	FUSE	HIGH VOLT BOARD LIGHT
1	1SCR/P1F	a, b, c	1	CR8 (T)
2	2SCR/P2F	a, b, c	2	CR9 (T)
3	3SCR/P3F	a, b, c	3	CR10 (T)
4	4SCR/N1R	b	1	CR10 (B)
5	5SCR/N2R	b	2	CR9 (B)
6	6SCR/N3R	b	3	CR8 (B)
4	4SCR/N1F	c	1	CR10(B)
5	5SCR/N2F	c	2	CR9 (B)
6	6SCR/N3F	c	3	CR8 (B)

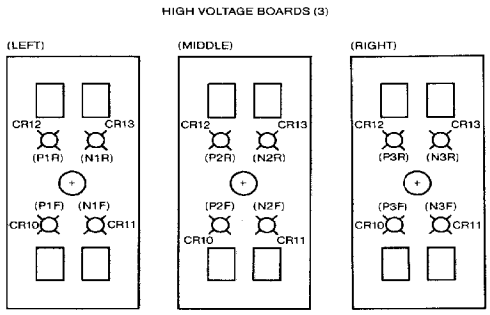
EXAMPLES:

1) CELL 4 SHORTED—CHECK CELL 4 (N1F) AND/OR
FUSE 1 IF CONFIGURATION c)
CHECK CELL 1/4 (P1F/N1R AND/OR
FUSE 1 IF IN CONFIGURATION b)

2) CELL 5 NOT FIRING—CHECK CELL 5(N2R) IF IN CONFIGURATION b)
CHECK CELL 5 (N2F) IF IN CONFIGURATION c)

MEDIUM HORSEPOWER





EXAMPLES:

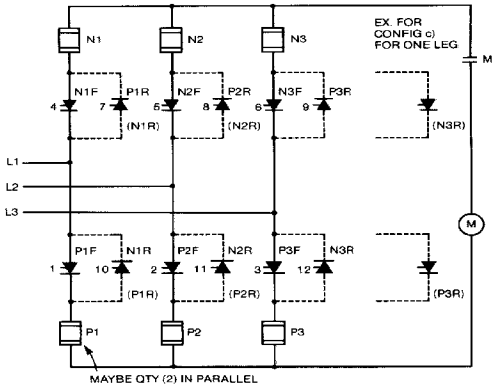
1) CELL 4 SHORTED—CHECK CELL 4 AND 7 (SRN1F/SRP1R)
AND/OR FUSE N1

2) CELL 5 NOT FIRING—CHECK CELL 5 (SRN2F) AND
OR FUSE N2

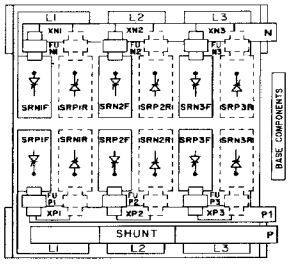
CONFIGURATION FOR MEDIUM AND LARGE HORSEPOWER

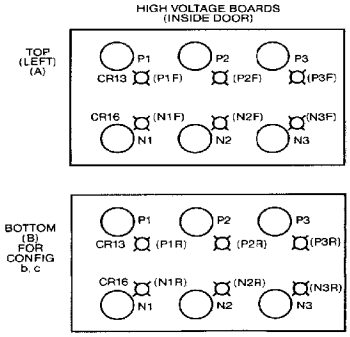
DOTTED LINE FOR REVERSING OR "SUPERBRIDGE"

a) NON REV. 6P
b) REV. 6P
c) NON REV. 6P ("SUPERBRIDGE")



LARGE HORSEPOWER





EXAMPLES:
SAME AS FOR
MEDIUM HORSEPOWER

CELL	NOMEN CLAT.	CONFIG	FUSE	HIGH VOLT. BOARD	
				MED	LARG
1	SR P1F	a, b	P1	CR10(L)	CR13(T)
2	SR P2F	a, b	P2	CR10(M)	CR14(T)
3	SR P3F	a, b	P3	CR10(R)	CR15(T)
4	SR N1F	a, b	N1	CR11(L)	CR16(T)
5	SR N2F	a, b	N2	CR11(M)	CR17(T)
6	SR N3F	a, b	N3	CR11(R)	CR18(T)
7	SR P1R	b, c	P1	CR12(L)	CR13(B)
8	SR P2R	b, c	P2	CR12(M)	CR14(B)
9	SR P3R	b, c	P3	CR12(R)	CR15(B)
10	SR N1R	b, c	N1	CR13(L)	CR16(B)
11	SR N2R	b, c	N2	CR13(M)	CR17(B)
12	SR N3R	b, c	N3	CR13(R)	CR18(B)

NOTE: IN CONFIGURATION c) "REVERSE" SCR's, P1R-N3R
ARE IN PARALLEL (ie. ANODE--ANODE) WITH "FORWARD"
SCR's P1F-N3F FOR HIGHER AMPERAGE. NOMENCLATURE
AS INDICATED IN PARENTHESIS ( )

ALSO PARALLEL CELLS (ie P1F/P1R) WILL BE FIRED SIMULTANEOUSLY